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A Transparent Financial Risk Factor Library

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Abstract

We introduce `tidyfinance`, an R package on CRAN for the core data workflow in empirical asset pricing: downloading stock returns and financial statements, constructing sorting variables, forming quantile portfolios, and computing long-short factor returns. A unified `download_data()` interface covers WRDS (CRSP, Compustat), Ken French's Data Library, FRED, and further sources. Modular functions handle fiscal-year lagging, breakpoint computation, and portfolio aggregation; every construction choice has a documented default and can be varied independently, making pipelines transparent and reproducible. We illustrate the `tidyfinance` workflow with a risk factor library covering 50 sorting variables across 841536 precomputed specifications. The library is accessible via an open-source API and an interactive dashboard at factors.tidy-finance.org.

Keywords: financial risk factors, reproducible research, asset pricing, API, R.

1. Introduction: Constructing financial risk factors

> **Note**
>
> Core contribution of the paper: `tidyfinance`, a modular, open-source library for reproducible workflows in empirical asset pricing. The relevant steps are a unified data access interface, modular functions for data processing, especially lagging, filtering and implementing portfolio sorts. A secondary contribution is the open-source factor library covering 50 sorting variables across 841536 specifications. The library is available via an API and an interactive dashboard at factors.tidy-finance.org.

- Python is not primary focus of the paper. We mention `py-tidyfinance` in the very end of the introduction to highlight access to the factor library and once when we show how to use `download_data`.

Factor returns drive empirical asset pricing tests, portfolio allocation, and performance attribution. Every series, however, embeds dozens of methodological choices: how breakpoints are set, whether portfolios are equal- or value-weighted, how delistings are handled, and which fiscal-year lag is applied. These choices are rarely standardized and almost never published with code. Researchers who reproduce or extend a factor series must reconstruct each pipeline from scratch, with no guarantee their choices match those of the original authors.

No widely adopted, transparent source standardizes factor construction. Current practice relies on informal publication of factor series without code or documented workflows, limiting reproducibility, comparability, and extension. Existing resources illustrate the gap: the French Data Library (Fama and French 2023) provides canonical factors but no code; the Open Source Asset Pricing Project (Chen and Zimmermann 2022) focuses on replication; the global factor database (Jensen, Kelly, and Pedersen 2023) targets Python only. No unified, open-source alternative exists.

We introduce `tidyfinance`, a modular framework (R on CRAN, Python on PyPI) for reproducible factor construction. The library covers 50 sorting variables and 841536 total specifications. We document and test every design choice (breakpoints, weighting, rebalancing, sample inclusion) and expose each as a configurable parameter.

The package implements an end-to-end pipeline: data acquisition from the Wharton Research Data Services (WRDS), Compustat, and other common providers; automated variable construction with standard lagging and cleaning routines; and portfolio aggregation with configurable sorting. A programmatic application programming interface (API) provides access to precomputed factor series with versioning and batch retrieval. Researchers using python access the same data download functions and factor library via `py-tidyfinance`. An interactive dashboard at factors.tidy-finance.org makes the library accessible without code. All software and workflows are publicly available to enable replication, sensitivity analysis, and extension.

Hou, Xue, and Zhang (2020) replicate 452 cross-sectional anomalies and find that most fail to survive out-of-sample, attributing failures in part to undocumented construction variation across studies. Chen and Zimmermann (2022) respond by publishing code for 212 anomaly portfolios, but their library targets a fixed, documented set rather than a reconfigurable pipeline. Jensen *et al.* (2023) provide a global factor database without an open construction workflow. `tidyfinance` fills the gap all three leave open: every construction decision is explicit, configurable, and version-controlled, so researchers can trace and vary any assumption systematically.

Among R packages for quantitative finance, **tidyquant** (Dancho and Vaughan 2026), **PerformanceAnalytics** (Peterson and Carl 2026), and **quantmod** (Ryan and Ulrich 2025) support time-series retrieval and portfolio performance analytics. None begins at raw accounting and return data, applies the cleaning and lagging conventions factor construction requires, or exposes construction choices as configurable parameters. `tidyfinance` provides this upstream pipeline with a consistent API across both languages. Hence, it enables researchers to access

standardized factor series without rebuilding data workflows in each environment.

2. Package overview

Note

This section introduces the design choices and workflow of `tidyfinance`.

- Installation
- Brief description of the workflow (data acquisition, data processing, portfolio sorting) in general. Illustration how `tidyfinance` offers support along these dimensions with a subsection on each part.
- Data access: Show which data sources are available via `download_data()`, with examples of how to call the function for each source. We mention `py-tidyfinance` here once to highlight that the same function is available in Python.
- Also highlight the standard cleaning conventions applied to CRSP, Compustat, TRACE data.
- Define what `date` means in `tidyfinance` conventions (i.e. end of month for monthly data, fiscal year-end for annual data, etc.)
- Data processing: Description and illustration of lagging with `add_lagged_columns()`, showing how the function implements the Fama-French convention for fiscal-year lagging.
- Data processing: Brief mention and explanation of other useful helper functions (`winsorize`, `trim`, `estimate_model`, `compute_rolling_value`)
- Portfolio sorting: Description and illustration of the methodology as such as well as the relevant functions (`compute_breakpoints()`, `assign_portfolio()`, `compute_portfolio_returns()`). Describe the use of `implement_portfolio_sort()` as a wrapper that implements the full pipeline from sorting variable to portfolio returns in one step.

2.1. Installation and setup

You can install the released version of `tidyfinance` from CRAN via:

```
install.packages("tidyfinance")
```

To install the development version of `tidyfinance` from GitHub, first install the **pak** package if needed, then install the development version:

```
pak::pak("tidy-finance/r-tidyfinance")
```

2.2. Design and workflow

`tidyfinance` organises the empirical asset pricing workflow into three stages: data acquisition, variable construction, and portfolio sorting. Each stage involves dozens of small decisions, such as universe filters, cleaning rules, rebalancing conventions, and breakpoint definitions. These decisions are rarely documented uniformly across studies, making replication difficult.

Figure 1 shows the pipeline and the package functions at each stage. The functions expose documented defaults that follow established conventions and configurable parameters where reasonable researchers might disagree. In this way, researchers can adopt the full pipeline or call individual functions within their own workflows.[1]

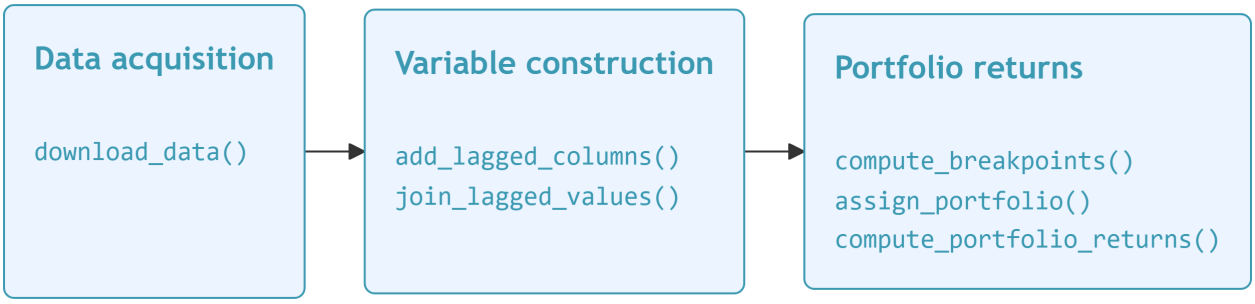


Figure 1: Research workflow for factor construction with `tidyfinance`, illustrating the three-stage pipeline and the package functions involved at each step.

2.3. Data access

`tidyfinance` provides a single interface, `download_data()`, to 320 dataset types: equity returns, fixed-income transactions, macroeconomic predictors, index constituents, and more. Table 1 lists all available sources. `list_supported_types()` and `list_supported_indexes()` return the full catalogue programmatically.

Source	Description	Example
Public data sources		
Ken French's Data Library	Fama-French 169+ factor models, portfolios, industry classifications	`download_data( "famafrench", "Fama/French 5 Factors")`
Global Q	Hou-Xue-Zhang Q5 factor model (market, size, investment, ROE) at multiple frequencies	`download_data( "globalq", "factors_q5_monthly")`
Goyal-Welch Predictors	Macroeconomic return predictors (DP, EP, BM, spreads, etc.)	`download_data( "macro_predictors", "monthly")`
FRED	St. Louis Fed economic time series (GDP, CPI, etc.)	`download_data( "fred", series = "GDP")`
Yahoo Finance	Daily OHLCV with adjusted close for any ticker	`download_data( "stock_prices", symbols = "AAPL")`

[1] A companion Python package `tidyfinance` provides the same interface with identical function signatures and dataset identifiers, so analyses can be ported between the two languages without changes to the data layer.

Source	Description	Example
Index Constituents	Current members of 21 major indexes (S&P 500, Russell, DAX, MSCI World, etc.)	`download_data( "constituents", index = "DAX")`
Open Source Asset Pricing	213+ monthly asset pricing signals and anomalies	`download_data("osap")`
WRDS (subscription required)		
CRSP	Stock returns, market cap, exchange, industry	`download_data( "wrds", "crsp_monthly")`
Compustat	Annual and quarterly financial statements	`download_data( "wrds", "compustat_annual")`
CCM Links	CRSP-Compustat linking table (permno to gvkey)	`download_data( "wrds", "ccm_links")`
FISD	Mergent Fixed Income Securities Database	`download_data("wrds", "fisd")`
TRACE	Corporate bond transactions (Dick-Nielsen cleaned)	`download_data( "wrds", "trace_enhanced", cusips = "655844CJ5")`

Table 1: Data sources available via `download_data()` in `tidyfinance`.

`download_data()` applies standard cleaning conventions automatically. For CRSP monthly data (version 2), the function restricts the universe to common equity securities traded on NYSE, AMEX, or NASDAQ. It computes market capitalisation as shares outstanding times price, constructs a one-month lag for portfolio weights, and maps exchange codes and SIC codes to readable labels. Excess returns are raw returns minus the Fama-French risk-free rate; observations with missing returns or market capitalisation are dropped. CRSP version 2 incorporates delisting returns directly, eliminating the need for a separate adjustment.

For Compustat annual data, `download_data()` restricts to industrial-format consolidated statements and retains only the most recent observation per firm-year. It constructs book equity through a coalesce hierarchy: stockholders' equity, then common equity plus preferred stock, then total assets minus total liabilities. The function adjusts for deferred taxes and investment tax credits and subtracts preferred stock.

All datasets returned by `download_data()` use a single `date` column that follows the same rules across frequencies. Daily observations return the trading date. Monthly observations are dated to the last calendar day of the reference month, regardless of whether that day was a trading day. Quarterly and annual observations take the fiscal period-end date as reported by the original source; for Compustat, this is the firm's fiscal year-end (`datadate`), which can differ from December across firms and across years for the same firm. Macroeconomic series from FRED keep their original release dates. A simple equality join on `date` therefore aligns observations of the same frequency, while cross-frequency joins, such as merging monthly

returns with annual fundamentals, require the lagging logic described in the next subsection.

2.4. Variable construction

Most predictive variables in asset pricing studies are based on lagged information so that portfolio formation in month t should always use only data observable at the end of month $t-1$. `add_lagged_columns()` automates this. For each row in the data, it looks back over a configurable window and attaches the most recent observed value of the chosen columns. By default (`ff_adjustment = FALSE`), the window is bounded by `lag` (the minimum delay between observation and use) and `max_lag` (the maximum delay before a stale value is treated as missing). Within this window, the function picks the most recent non-missing value; if no observation falls in the window, the lagged column takes `NA`.

A second mode, enabled by `ff_adjustment = TRUE`, implements the Fama-French convention (Fama and French 1993). Accounting variables for fiscal year $t-1$ are assumed to become public information by June of year t and are then used for portfolio formation from July of t through June of $t+1$. This produces step-function lagged values that update once per year at the June/July boundary, regardless of the underlying fiscal year-end. Figure 2 contrasts the two mechanisms. The upper panel shows a standard look-back window controlled by `lag` and `max_lag`. The lower panel implements the Fama-French convention: fiscal-year data from year $t-1$ becomes available only from July of year t, so all months from July through June share the same lagged value locked in at the previous rebalancing. In both cases, `add_lagged_columns()` selects the most recent observed value within the window.

`join_lagged_values()` is a related helper that merges lagged values from one data frame into another using the same windowing logic. It is useful when the target panel and the source panel are at different frequencies, for example when joining monthly accounting data into a quarterly target panel. The function takes the same `ff_adjustment` argument as `add_lagged_columns()`, so the Fama-French convention can be applied at the join step rather than as a separate transformation.

A handful of additional helpers cover routine operations that recur across empirical studies. `winsorize()` and `trim()` handle outliers in cross-sectional variables: the former replaces values beyond user-specified quantiles with the quantile values themselves, while the latter sets them to `NA`. `estimate_model()` fits a linear model from a formula string and returns the coefficients as a tidy one-row data frame, which composes naturally with `group_by()` for grouped or rolling regressions such as time-series factor exposures. `compute_rolling_value()` applies an arbitrary aggregation function over a rolling window defined in calendar time rather than row counts, accommodating irregular observation frequencies and missing periods.

2.5. Portfolio sorting

Portfolio sorts are among the most widely used tools in empirical asset pricing (Bali, Engle, and Murray 2016). The procedure groups stocks each period into portfolios based on a sorting variable, such as size, beta, book-to-market, or momentum, and then compares return distributions across the resulting portfolios. The long-short return between the extreme portfolios is interpreted as a factor-mimicking strategy and serves as the basis for tests of

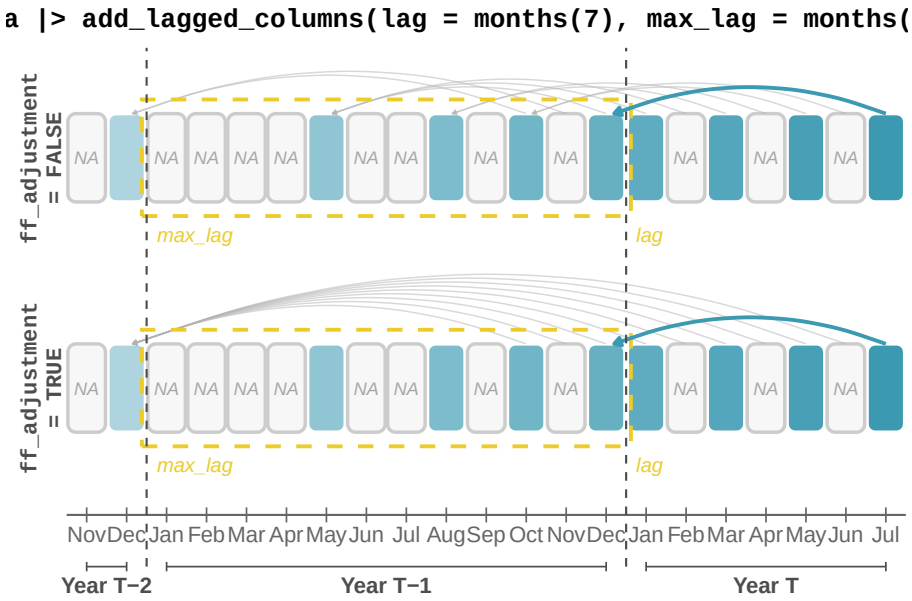


Figure 2: Two lagging mechanisms in `add_lagged_columns()`. Each row represents a data series with observed (filled) and missing (hollow) cells. Dashed rectangles mark the search window; arrows show which value gets matched to each target month.

whether the sorting variable forecasts future returns. We expose the procedure through four composable functions.

`compute_breakpoints()` takes a cross-section and a sorting variable and returns the numeric cutoffs that define the portfolio boundaries. The user supplies either a number of portfolios (for example, five for quintiles) or an explicit vector of percentiles, but not both. A `breakpoints_exchanges` argument restricts the universe used to compute breakpoints. The standard convention is NYSE-only, which prevents the large cross-section of small NASDAQ firms from dominating the lower bins. A related `breakpoints_min_size_threshold` argument excludes stocks below a market-cap quantile from the breakpoint computation, which is useful for size-related sorts where microcaps would otherwise distort the cutoffs. A `smooth_bunching` flag handles cases where the sorting variable clusters at the edges by pinning extreme values to the boundary portfolios and recomputing breakpoints on the remainder.

`assign_portfolio()` maps each observation to a portfolio integer using the breakpoints. The breakpoint computation is delegated to a user-supplied function (defaulting to `compute_breakpoints()`), so custom rules such as dollar-based cutoffs or industry-adjusted breakpoints can be plugged in without rewriting the assignment logic. In practice, `assign_portfolio()` is called separately for each rebalancing date, so that portfolios are reformed each period rather than fixed once for the whole sample.

`compute_portfolio_returns()` ties the lower-level pieces together into a single call. Given a panel of stock data, one or two sorting variables, a sorting method, and `breakpoint_options`, it handles rebalancing, breakpoint computation, portfolio assignment, and return aggregation. It then returns a complete panel of portfolio returns. For each portfolio-date the function

reports three return measures: equal-weighted, value-weighted using `mktcap_lag` as the weight, and capped value-weighted. The capped version pulls weights above the `cap_weight` percentile (default 0.8) back to that cap, which prevents a small number of megacap firms from dominating the portfolio. A `min_portfolio_size` argument sets the minimum number of constituents below which the corresponding return is set to `NA`. An optional `rebalancing_month` argument switches from period-by-period rebalancing to annual rebalancing in the specified month. Portfolios formed in that month are then held constant for the following twelve months.

The function supports three sorting schemes, illustrated in Figure 3. A *univariate* sort uses a single variable. An *independent bivariate* sort computes breakpoints on each variable separately and intersects the resulting bins to produce a two-way grid. A *dependent bivariate* sort first partitions the cross-section by the secondary (control) variable and then re-sorts within each bucket on the main variable, which is the appropriate choice when the second variable is treated as a control rather than as an independent dimension of variation. For bivariate sorts, returns are averaged over the secondary variable so that only the main-variable portfolios are returned.

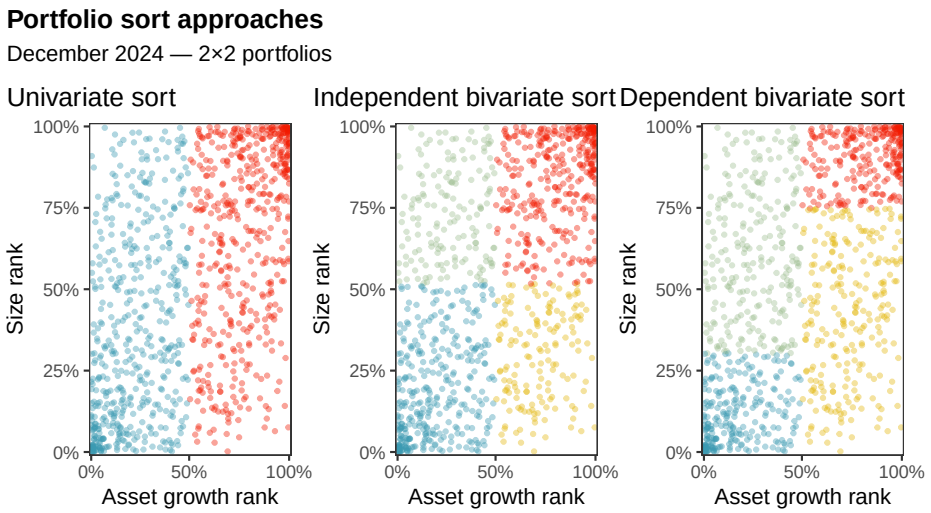


Figure 3: Portfolio sort approaches illustrated for a single month (December 2024). Each panel plots stock-month observations assigned to `r .QuartoInlineRender(n_portfolio)`×`r .QuartoInlineRender(n_portfolio)` portfolios based on asset growth rank (x-axis) and size rank (y-axis), both expressed as percentiles. Colors indicate portfolio cell assignment. The univariate sort (left) assigns portfolios on asset growth alone; the independent bivariate sort (center) assigns each dimension independently; the dependent bivariate sort (right) sorts on size first, then on asset growth within size groups.

`compute_long_short_returns()` is a thin convenience layer that takes the output of `compute_portfolio_returns()` and computes the spread between the top and bottom portfolios. A `direction` argument controls the sign and defaults to `"top_minus_bottom"`. The function returns one long-short series per return measure available in the input, so the equal-weighted, value-weighted, and capped value-weighted spreads are produced in a single call. This is the long-short factor-mimicking return that is usually the object of interest in the asset-pricing literature.

3. Vignette: A case study in factor construction

> **Note**
>
> This section shows the workflow from scratch without excessive documentation by going through data aggregation, variable construction for asset growth, lagging and sorting into portfolios. The purpose is to illustrate the workflow and the relevant functions in a concrete example. We do not show every step of the data cleaning process for brevity.
>
> - Compute asset growth risk factor time series once with defaults (describe the defaults) using `implement_portfolio_sort()`.
> - use the `filter_options`, `breakpoint_options`, and other arguments to streamline the process for an *exotic* example.
> - Illustrate "standard" summary stats for one specification of the factor or use fama-mac beth functionality for some standard testing.

We walk through the construction of the asset growth factor from raw CRSP and Compustat data to long-short portfolio returns, implementing each stage of Figure 1 in turn. Each step uses a `tidyfinance` function and generalises directly to any sorting variable in the factor library.[2]

Once installed, load `tidyfinance` and a minimal set of packages for data manipulation and visualisation.

```
library("tidyfinance")
```

```
library("dplyr")
library("tidyr")
library("lubridate")
```

3.1. Data download

Before the first WRDS download, call `set_wrds_credentials()` to store WRDS credentials securely in `.Renviron`; subsequent `download_data()` calls establish the database connection automatically.

```
date_start <- as.Date("1960-01-01")
date_end <- as.Date("2024-12-31")
```

First, we download monthly stock returns from CRSP. Setting `add_ccm_links = TRUE` appends CRSP-Compustat linking identifiers needed to merge with financial statement data in the next step.

```
crsp_monthly <- download_data(
  domain = "wrds",
  dataset = "crsp_monthly",
  version = "v2",
  start_date = date_start,
  end_date = date_end,
```

[2]Installation of the `tidyfinance` package is described in the appendix in Section 2.1 Installation and setup.

```
  add_ccm_links = TRUE
)
```

The dataset contains 3451594 stock-month observations for 26263 unique stocks, covering the period from 1960-01-01 to 2024-12-01.

Next, we download annual financial statements from Compustat.

```
compustat_annual <- download_data(
  domain = "wrds",
  dataset = "compustat_annual",
  start_date = date_start,
  end_date = date_end,
  additional_columns = c("at", "ib"),
  only_us = TRUE
)
```

Annual Compustat data contains 350813 firm-year observations for 22612 unique firms, covering the period from 1960-01-01 to 2024-12-01. The `additional_columns` argument retrieves total assets (`at`) and income before extraordinary items (`ib`), needed for computing asset growth and applying an earnings filter in later steps. Setting `only_us = TRUE` restricts the dataset to USD-denominated observations, ensuring consistency with CRSP.

3.2. Data cleaning and variable construction

We illustrate data cleaning and variable construction using asset growth (AG) as our running example. Asset growth is the year-over-year percentage change in total assets (Cooper, Gulen, and Schill 2008),

$$AG_t = (AT_t - AT_{t-1})/AT_{t-1}.$$

The `add_lagged_columns()` function from `tidyfinance` handles temporal alignment by creating a lagged version of total assets grouped by firm identifier (`gvkey`) with a one-year lag. Non-finite values (arising when lagged total assets are zero or missing) become `NA`.[3]

```
sorting_variable_data <- compustat_annual |>
  add_lagged_columns(
    "at",
    lag = years(1),
    by = "gvkey",
  ) |>
  mutate(
    asset_growth = (at - at_lag) / at_lag,
```

[3]For the sake of simplicity we do not show every step of the data cleaning process in the paper. For instance, we apply an earnings filter to exclude firms with negative or zero income before extraordinary items (`ib`), which is a common practice in the literature to mitigate the influence of financial distress on asset growth portfolios. This filter is implemented in the code but not shown in the paper for brevity. We also apply an age filter to exclude firms younger than two years, which is common in the literature to ensure that the sorting variable is based on stable financial information. This filter is also implemented in the code but not shown in the paper for brevity. The factor library API provides preprocessed panels with all these filters applied, so users can access the final sorting variables without having to implement these steps themselves.

```
    asset_growth = if_else(is.finite(asset_growth), asset_growth, NA_real_)
  ) |>
  select(gvkey, date, asset_growth)
```

`sorting_variable_data` contains one observation per firm-year, the natural unit for cross-sectional analysis of annual accounting variables. The package provides `filter_out()` as a convenience wrapper that removes rows matching a logical predicate, and `winsorize()` to clip numeric vectors at specified quantile bounds. Asset growth is heavily right-skewed: a small number of firms with dramatic balance-sheet expansions generate extreme positive values. We winsorize at the 1st and 99th percentiles before computing summary statistics. Table 2 reports the pooled cross-sectional summary; Table 3 shows how those moments vary across calendar years.

N	Mean	SD	P10	P25	Median	P75	P90
303671	0.235	0.786	-0.181	-0.028	0.071	0.216	0.596

Table 2: Cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). The table reports the number of firm-year observations (N), the mean, standard deviation (SD), and selected percentiles of the annual asset growth distribution computed from `sorting_variable_data`.

Years	Avg N	Mean (avg)	Mean (min)	Mean (max)	SD (avg)	Median (avg)	Median (min)	Median (max)
64	4745	0.239	0.042	0.904	0.788	0.072	0.001	0.134

Table 3: Time-series averages of annual cross-sectional summary statistics for asset growth (winsorized at the 1st and 99th percentiles). For each calendar year, cross-sectional mean, standard deviation, and median are computed from `sorting_variable_data`; the table reports the time-series mean, minimum, and maximum of those annual statistics.

The next step merges annual sorting variables onto the monthly CRSP stock panel using `join_lagged_values()`. The function aligns fiscal-year-end data with monthly observations through a configurable lag window. Setting `ff_adjustment = TRUE` implements the Fama-French convention (Fama and French 1993), aligning fiscal-year data from year t with portfolio formation in July of year $t+1$ through a 12-to-18-month look-back window. The same lagging process generalizes to all sorting variables in the full factor library. The factor library API provides precomputed lagged panels for all variables.

With both lagging mechanisms documented, we apply `join_lagged_values()` to merge the sorting variable onto the monthly return panel. Figure 4 shows the 33rd, 50th, and 66th percentiles of asset growth over time in the resulting panel; pronounced spikes during economic expansions and contractions highlight the cyclical nature of corporate investment that portfolio sorts aim to exploit.

```
sorting_data <- crsp_monthly |>
  join_lagged_values(
```

```
    sorting_variable_data,
    "gvkey",
    min_lag = months(7),
    max_lag = months(18),
    ff_adjustment = TRUE
  )

sorting_data <- sorting_data |>
  select(date, permno, exchange, ret_excess, mktcap_lag, asset_growth) |>
  filter(date > date_start + months(12))
```

The filter removes the initial year used to compute the growth rate to avoid mechanically introduced gaps from the lagging step, leaving a panel of stock-month observations with excess returns, exchange membership, lagged market capitalization, and the lagged sorting variable.

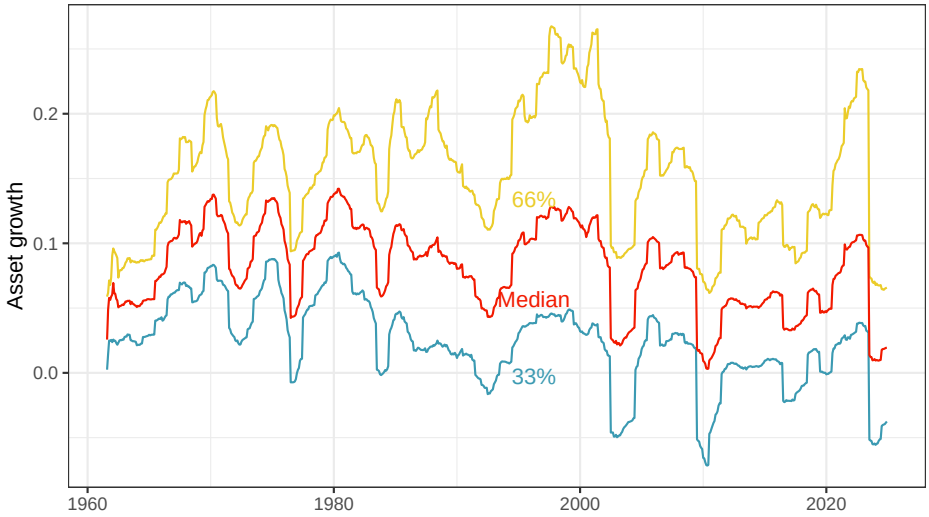


Figure 4: Distribution of asset growth over time. Lines show the 33rd, 50th, and 66th percentiles of the cross-sectional distribution of asset growth in `sorting_data`, the firm-month panel used as input for portfolio construction. Asset growth is expressed as a proportion (e.g., 0.2 = 20% annual growth in total assets) and covers the full sample period of the merged CRSP-Compustat dataset.

With the sorting variable constructed and merged onto the monthly stock panel, we now turn to portfolio sorts, the mechanism that translates firm-level characteristics into tradable factor returns.

3.3. Portfolio sorts

In portfolio sorts, stocks are ranked each period by their sorting variable and divided into quantile portfolios. Portfolio returns are equal-weighted or value-weighted averages of constituent stock returns. The long-short factor return is the difference between the extreme portfolios, for instance bottom minus top for asset growth, reflecting the negative premium documented by Cooper *et al.* (2008). Key methodological decisions include the number of

portfolios, the exchange universe for computing breakpoints, the weighting scheme, and the rebalancing frequency.

The `tidyfinance` package implements this workflow through three functions. First, `breakpoint_options()` configures the sorting parameters: the number of quantile portfolios and which exchanges determine the breakpoints. Second, `compute_portfolio_returns()` takes the sorting data, applies the specified configuration, and returns portfolio-level returns. Third, `compute_long_short_returns()` constructs the hedge portfolio return from the extreme quantiles.

Univariate sorts

`compute_portfolio_returns()` is built on two lower-level functions that are also available for custom workflows. `compute_breakpoints()` computes quantile cutpoints for the sorting variable from a specified exchange universe. `assign_portfolio()` maps each stock-month to a portfolio number based on those cutpoints. Researchers who need finer control, for instance to build overlapping portfolios, apply non-standard cutoffs, or weight within cells differently, can call these functions directly rather than through the high-level wrapper.

In a univariate sort, stocks are ranked on a single characteristic. We illustrate this with asset growth, forming decile portfolios using NYSE breakpoints and value-weighted returns, rebalanced annually in July.

```
options_main <- breakpoint_options(
  n_portfolios = 5,
  breakpoint_exchanges = "NYSE"
)
```

The `breakpoint_options()` call specifies that breakpoints come from NYSE-listed stocks only, a common convention that avoids the influence of small AMEX and NASDAQ firms on portfolio cutoffs. We then pass these options to `compute_portfolio_returns()`.

```
portfolio_returns <- compute_portfolio_returns(
  sorting_data,
  sorting_variables = "asset_growth",
  sorting_method = "univariate",
  rebalancing_month = 7,
  options_main
)
```

```
Returning a complete panel with 60 missing values in factor returns due to
insufficient observations (n() <= 0).
```

`compute_portfolio_returns()` returns a long tibble with one row per portfolio-month, containing value-weighted (`ret_excess_vw`), equal-weighted (`ret_excess_ew`), and capped value-weighted (`ret_excess_vw_capped`) excess returns. The long-short factor return is obtained via `compute_long_short_returns()`. The `direction = "bottom_minus_top"` argument means the factor is long low-AG stocks and short high-AG stocks, consistent with the negative premium documented by Cooper *et al.* (2008).

```
factor_returns_ag <- portfolio_returns |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Table 4 summarises the time series of each quintile across all three weighting schemes, with the long-short portfolio appended as the final row.

	VW		EW		VWC	
Portfolio	Mean	VW t-stat	Mean	EW t-stat	Mean	VWC t-stat
1	0.095	4.338	0.156	4.876	0.107	4.065
2	0.079	4.295	0.122	5.241	0.097	4.543
3	0.074	4.005	0.109	4.893	0.093	4.484
4	0.073	3.392	0.093	3.740	0.085	3.644
5	0.063	2.369	0.036	1.114	0.041	1.372
L/S	0.033	2.286	0.120	9.370	0.066	5.167

Table 4: Time-series averages of annualised monthly excess returns for asset growth quintile portfolios. Portfolios are formed annually in July using NYSE breakpoints, rebalanced annually. Means are annualised (multiplied by 12). The t~statistic tests whether the mean monthly excess return differs from zero. The L/S row is the long-short portfolio (bottom minus top quintile).

The value-weighted long-short portfolio earns 3.3% per year (t~statistic 2.29), consistent with the negative premium documented by Cooper *et al.* (2008). The equal-weighted spread is larger, as expected for characteristics that correlate with size. The monotone decline across quintiles confirms that `tidyfinance` reproduces the literature result from raw WRDS data.

Bivariate sorts

Bivariate (double) sorts control for a second variable, typically firm size (`mktcap_lag`), when forming portfolios. In dependent sorts, stocks are first sorted into groups by the secondary variable; within each group, they are then sorted on the main characteristic. In independent sorts, both sorts apply unconditionally, and portfolios form at the intersections. The factor return averages across the secondary dimension, isolating the effect of the primary variable after controlling for size. Figure 3 illustrates the three approaches for a single cross-section.

We reuse `options_main` from the univariate sort (five portfolios, NYSE breakpoints) and add a separate options object for the secondary size dimension:

```
options_secondary <- breakpoint_options(
  n_portfolios = 2,
  breakpoint_exchanges = "NYSE"
)

portfolio_returns_biv <- compute_portfolio_returns(
  sorting_data,
  c("asset_growth", "mktcap_lag"),
  sorting_method = "bivariate-dependent",
```

```
  rebalancing_month = 7,
  options_main,
  options_secondary
)
```

```
Returning a complete panel with 60 missing values in factor returns due to
insufficient observations (n() <= 0).
```

```
factor_returns_biv <- portfolio_returns_biv |>
  compute_long_short_returns(direction = "bottom_minus_top")
```

Sort	VW Mean	VW t~stat
Univariate	0.033	2.286
Bivariate (size-controlled)	0.060	5.046

Table 5: Univariate vs size-controlled long-short returns for asset growth. Annualized value-weighted mean excess return and t~statistic for the univariate sort and the dependent bivariate sort averaging across two size groups. Portfolios use NYSE breakpoints and annual July rebalancing.

The AG premium survives size control, suggesting it captures an investment effect distinct from the size factor. The `sorting_method` argument switches between dependent and independent double sorts, so researchers can inspect both configurations without modifying the pipeline.

4. Factor library and API

Note

This section introduces the full factor library and API.

- Explain the computation of the entire grid of factor returns across all combinations of methodological choices. Cite WWW and Gruber et al. for the motivation of this comprehensive approach.
- Emphasise that the library is available for download, so researchers can access results without re-running the computationally intensive portfolio sort pipeline.
- Show how to download one default time series and how to use `fill_all`.
- Illustrate Sharpe-ratio variation.
- Mention that ´py-tidyfinance‘ can be used with identical callss

4.1. Specification grid

The examples above represent single points in a high-dimensional space of methodological choices. Table 6 summarises the preprocessing decisions that can be varied when constructing factor returns. The full factor library exhaustively evaluates all valid combinations of these choices for each of the sorting variables listed in Table 7.

Preprocessing choice	Argument	Options	Description
Size exclusion	`exclude_size`	None; *bottom 20%*	Fraction of the smallest stocks (by NYSE market cap) excluded from the universe.
Exclusion of financials	`exclude_financials`	*No*; Yes	Drops financial-sector stocks (SIC 6000–6999) from the universe.
Exclusion of utilities	`exclude_utilities`	*No*; Yes	Drops utility-sector stocks (SIC 4900–4999) from the universe.
Exclusion of neg. earnings	`exclude_negative_earnings`	*No*; Yes	Drops firms with negative earnings before sorting.
Sorting variable lag	`sorting_variable_lag`	3m; *6m*; FF	Lag applied to the sorting variable before portfolio assignment.
Rebalancing	`rebalancing`	*Monthly*; Annual	How frequently portfolios are reformed.
BP: Quantiles (main)	`breakpoints_main`	3; 5; *10*	Number of quantile groups for the portfolio sort.
Sort method	`sorting_method`	*Single*; Double dependent; Double independent	Single sort or sequential double sort (by size).
BP: Quantiles (secondary)	`breakpoints_secondary`	*Ind*; 2; 5	Groups for the secondary sort variable; only relevant for double sorts.
BP: Exchanges	`breakpoints_exchanges`	*NYSE*; All	Exchange(s) used to compute quantile breakpoints.
Weighting scheme	`weighting_scheme`	EW; *VW*; truncated VW	Return weighting within portfolios.

Table 6: Preprocessing choices in the portfolio sort grid and corresponding `download_data()` arguments. Italic values indicate the default. For double sorts, the secondary quantiles and exchange choices apply to the size dimension only; for univariate sorts, those rows are not applicable. FF refers to the lag procedure of Fama and French (1993).

We host precomputed factor returns for all specifications on Hugging Face and make them accessible via the `tidyfinance` API. Researchers can retrieve results without re-running the computationally intensive portfolio sort pipeline. An interactive download interface is also available at factors.tidy-finance.org.

The factor library covers 50 sorting variables across seven categories: investment, profitability, valuation, momentum, intangibles, trading frictions, and financing. The selection follows Jensen *et al.* (2023). Table 7 lists all variables with their long-short direction and primary reference.

4.2. The factor library

This sensitivity is not specific to AG. For all 50 sorting variables in the library, construction choices shift Sharpe ratios by enough to reverse the sign of the premium for some factors. Figure 5 shows the distribution of annualized Sharpe ratios for each variable across all 841536 specifications.

Figure 5 conveys two findings. First, most distributions are wide: construction choices have material effects on factor performance across the library. Second, for several factors the distribution spans zero, meaning the premium's sign depends on the specification. A researcher reporting a single undocumented series risks mistaking a construction artifact for an economic signal. Factors with narrow distributions centered well above zero are construction-robust; those with wide or zero-straddling distributions require sensitivity analysis before drawing conclusions.

Table 7 lists all 50 sorting variables with their long-short direction and primary reference.

Variable	Name		Main reference
52w	52-Week High	↑	George and Hwang (2004)
aci	Abnormal Corporate Investment	↓	Titman, Wei, and Xie (2004)
ami	Amihud Illiquidity	↑	Amihud (2002)
ag	Asset Growth	↓	Cooper *et al.* (2008)
beta	Beta	↑	Fama and MacBeth (1973)
bm	Book-to-Market	↑	Daniel and Titman (2006)
cfm	Cash Flow to Market	↑	Lakonishok, Shleifer, and Vishny (1994)
cbop	Cash-Based Operating Profitability	↑	Ball, Gerakos, Linnainmaa, and Nikolaev (2016)
dbe	Change in Common Equity	↓	Richardson, Sloan, Soliman, and Tuna (2005)
dcol	Change in Current Operating Liabilities	↓	Richardson *et al.* (2005)
dfnl	Change in Financial Liabilities	↓	Richardson *et al.* (2005)
dnoa	Change in Net Operating Assets	↓	Hirshleifer, Hou, Teoh, and Zhang (2004)
dpia	Change in PPE and Intangibles	↓	Lyandres, Sun, and Zhang (2008)
dwc	Change in Working Capital	↓	Richardson *et al.* (2005)
cdi	Composite Debt Issuance	↓	Lyandres *et al.* (2008)
csi	Composite Share Issuance	↓	Daniel and Titman (2006)
dm	Debt to Market	↑	Bhandari (1988)
dtv	Dollar Trading Volume	↓	Brennan, Chordia, and Subrahmanyam (1998)
em	Earnings to Market	↑	Basu (1983)
ebm	Enterprise Book to Market	↑	Penman, Richardson, and Tuna (2007)
gpa	Gross Profits to Assets	↑	Novy-Marx (2013)
hr	Hiring Rate	↓	Belo, Lin, and Bazdresch (2014)
iskew	Idiosyncratic Skewness	↓	Bali *et al.* (2016)

ivol	Idiosyncratic Volatility	↓	Ang, Hodrick, Xing, and Zhang (2006)
dinv	Inventory Changes	↓	Thomas and Zhang (2002)
ig	Investment Growth	↓	Xing (2008)
rev	Long-Term Reversal	↓	De Bondt and Thaler (1985)
me	Market Equity	↓	Banz (1981)
mdr	Maximum Daily Return	↓	Bali, Cakici, and Whitelaw (2011)
mom	Momentum	↑	Fama and French (1996)
ndf	Net Debt Financing	↓	Baker and Wurgler (2006)
ndm	Net Debt to Market	↓	Penman *et al.* (2007)
nef	Net Equity Financing	↓	Baker and Wurgler (2006)
nxf	Net External Financing	↓	Baker and Wurgler (2006)
noa	Net Operating Assets	↓	Hirshleifer *et al.* (2004)
npy	Net Payout Yield	↑	Boudoukh, Michaely, Richardson, and Roberts (2007)
o	O-Score	↓	Ohlson (1980); Dichev (1998)
oa	Operating Accruals	↓	Sloan (1996)
ocm	Operating Cash Flow to Market	↑	Desai, Rajgopal, and Venkatachalam (2004)
ope	Operating Efficiency	↑	Fama and French (2015)
ol	Operating Leverage	↑	Novy-Marx (2011)
rer	Real-Estate Ratio	↑	Titman *et al.* (2004)
rs	Relative Strength	↑	Jegadeesh and Livnat (2006)
rmom	Residual Momentum	↑	Blitz, Huij, and Martens (2011)
roa	Return on Assets	↑	Balakrishnan, Bartov, and Faurel (2010)
roe	Return on Equity	↑	Hou, Xue, and Zhang (2014)
sm	Sales to Market	↑	Barbee Jr, Mukherji, and Raines (1996)
tur	Share Turnover	↓	Datar, Naik, and Radcliffe (1998)
sue	Standardized Unexpected Earnings	↑	Foster, Olsen, and Shevlin (1984)
tbi	Taxable Income to Book Income	↓	Lev and Nissim (2004)

Table 7: Sorting variables in the factor library. The direction column indicates whether the long-short factor return goes long top-minus-bottom (↑) or bottom-minus-top (↓) of the characteristic distribution.

To download factor returns from the factor library, call

```
asset_growth_returns <- download_data(
  "tidyfinance",
  "factor_library",
  sorting_variable = "ag",
  sorting_variable_lag = "6m"
)
```

The defaults passed to `dataset = "factor_library"` reflect one common construction choice

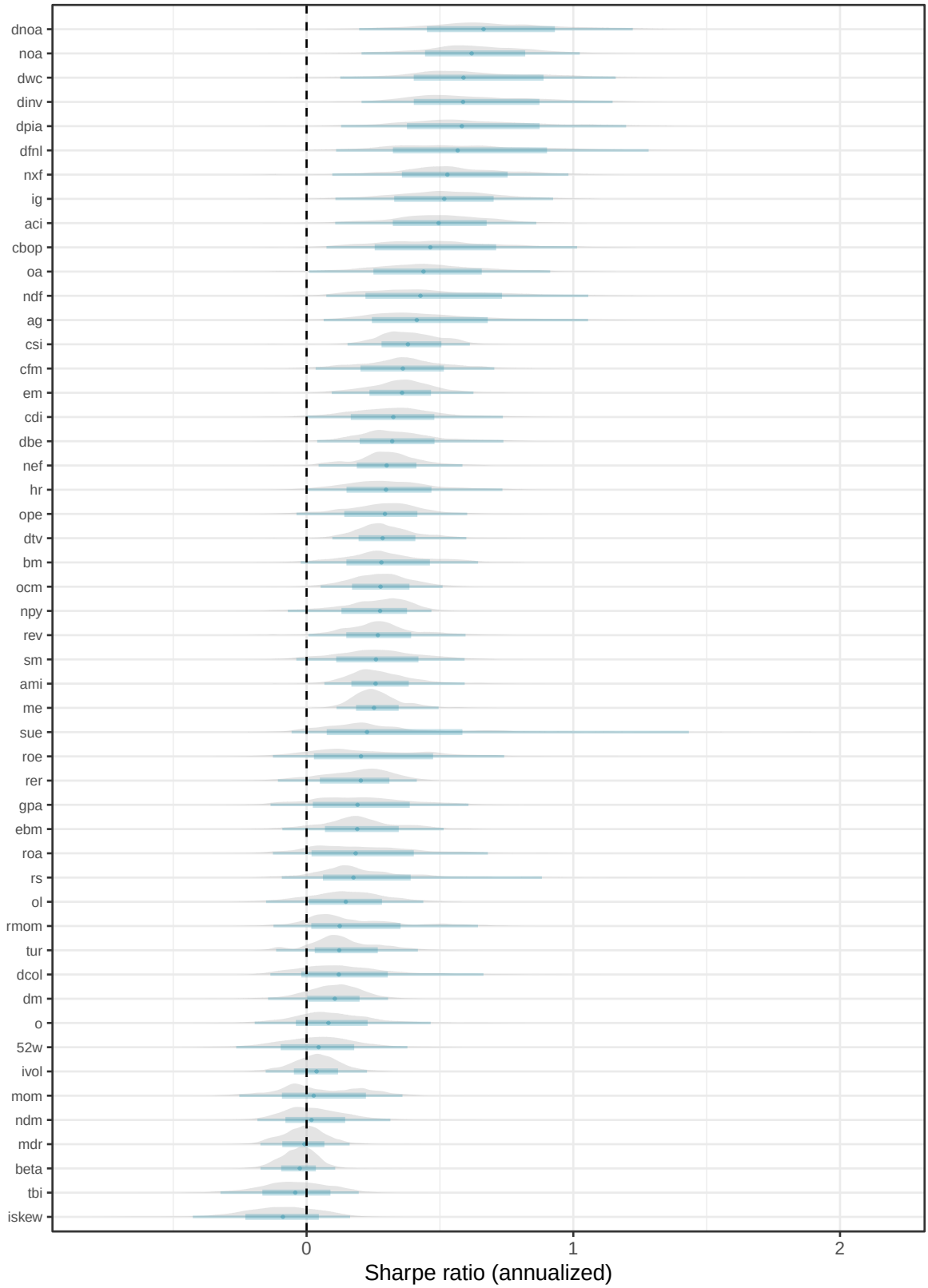


Figure 5: Sharpe ratios across construction choices. Ridge plots show the distribution of annualized Sharpe ratios of long-short return differentials for each sorting variable across all `r .QuartoInlineRender(nrow(portfolio_sort_grid))` specifications illustrated in Table 6, based on the available sample period per factor. Most variables exhibit substantial variation across specifications, underscoring the sensitivity of factor returns to construction choices.

but may not suit every research question. Always verify that the selected combination matches the intended design before drawing conclusions.

```
download_data(
  "tidyfinance",
  "factor_library",
  sorting_variable = "52w",
  sorting_variable_lag = "6m",
  fill_all = TRUE
)
```

For finer control, `tidyfinance` exposes lower-level helpers that sit beneath the `download_data()` interface. `get_available_huggingface_files()` returns all available files for a given Hugging Face repository as a tibble; `download_factor_library_returns_ids()` retrieves returns for a specific vector of precomputed specification IDs. Researchers who want to iterate over a custom subset of specifications rather than filter by parameter value will prefer this route.

```
full_grid <- get_available_huggingface_files(
  "tidy-finance",
  "factor-library-grid"
) |>
  dplyr::pull(.data$url) |>
  arrow::read_parquet()

tidyfinance:::download_factor_library_ids(123)
```

5. Summary and discussion

> **Note**
>
> Keep brief or delete entirely, as the main paper's conclusion already covers the key points.

Factor returns encode their construction history. Two researchers following the same paper can produce different factor series, with no way of knowing why, because breakpoints, weighting, and lagging conventions are rarely published alongside the data. `tidyfinance` makes these decisions visible: the package covers 50 sorting variables across 841536 documented specifications, each reproducible from raw WRDS and Compustat data in a single, version-controlled workflow.

The library currently covers US equities. Construction-choice sensitivity in international markets is beyond its current scope.

The broader implication is methodological. Researchers who disagree about a construction choice can resolve the disagreement empirically: run both variants through the same pipeline, compare factor returns, and report the difference. The API and interactive dashboard extend this accessibility to practitioners who need standardized series without running the full pipeline.

With 841536 specifications across 50 sorting variables, the library makes systematic sensitivity analysis newly tractable. This opens a path toward meta-analytic evidence on which factor

properties are robust and which are artifacts of undocumented defaults.

Computational details

We obtained the results in this paper using R~4.5.2 with the following packages: **tidyfinance**~0.4.5.9027, **dplyr**~1.2.0, **tidyr**~1.3.1, **ggplot2**~4.0.0, **lubridate**~1.9.4, **arrow**~21.0.0.1, **stringr**~1.5.2, **forcats**~1.0.0, **ggdist**~3.3.3, and **knitr**~1.51. CRAN hosts R and all packages at https://CRAN.R-project.org/. `renv`~1.1.5 ensures full reproducibility of the computational environment. Reproducing the factor return data requires an institutional subscription to Wharton Research Data Services (WRDS), as the underlying CRSP and Compustat data are proprietary and cannot be redistributed.

We publish all code, data workflows, and documentation at https://github.com/tidy-finance/jss-multilingual-factor-library. The code is released under the MIT License, which is compatible with the GNU General Public License (GPL).

```
sessionInfo()
```

```
R version 4.5.2 (2025-10-31 ucrt)
Platform: x86_64-w64-mingw32/x64
Running under: Windows 11 x64 (build 26200)

Matrix products: default
  LAPACK version 3.12.1

locale:
[1] LC_COLLATE=English_United States.utf8
[2] LC_CTYPE=English_United States.utf8
[3] LC_MONETARY=English_United States.utf8
[4] LC_NUMERIC=C
[5] LC_TIME=English_United States.utf8

time zone: Europe/Copenhagen
tzcode source: internal

attached base packages:
[1] stats     graphics  grDevices datasets  utils     methods   base

other attached packages:
[1] patchwork_1.3.2        ggforce_0.5.0          ggplot2_4.0.0
[4] lubridate_1.9.4        tidyr_1.3.1            dplyr_1.2.0
[7] tidyfinance_0.4.5.9027

loaded via a namespace (and not attached):
 [1] ggdist_3.3.3         sandwich_3.1-1       rappdirs_0.3.3
 [4] generics_0.1.4       renv_1.1.5           stringi_1.8.7
 [7] lattice_0.22-7       hms_1.1.4            digest_0.6.37
```

```
[10] magrittr_2.0.4       RColorBrewer_1.1-3   evaluate_1.0.5
[13] grid_4.5.2           timechange_0.3.0     fastmap_1.2.0
[16] blob_1.3.0           jsonlite_2.0.0       RPostgres_1.4.9
[19] DBI_1.2.3            purrr_1.1.0          scales_1.4.0
[22] frenchdata_0.2.0     tweenr_2.0.3         httr2_1.2.2
[25] cli_3.6.5            rlang_1.1.7          polyclip_1.10-7
[28] dbplyr_2.5.2.9000    bit64_4.6.0-1        withr_3.0.2
[31] yaml_2.3.10          tools_4.5.2          tzdb_0.5.0
[34] curl_7.0.0           assertthat_0.2.1     vctrs_0.7.2
[37] R6_2.6.1             zoo_1.8-15           lifecycle_1.0.5
[40] stringr_1.5.2        bit_4.6.0            MASS_7.3-65
[43] arrow_21.0.0.1       pkgconfig_2.0.3      pillar_1.11.1
[46] gtable_0.3.6         Rcpp_1.1.0           glue_1.8.0
[49] xfun_0.53            tibble_3.3.0         tidyselect_1.2.1
[52] knitr_1.51           farver_2.1.2         htmltools_0.5.8.1
[55] labeling_0.4.3       rmarkdown_2.29       compiler_4.5.2
[58] S7_0.2.0             distributional_0.7.0
```

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Internet Appendix

IA-1. Variable construction

In this section, we show the construction details for the risk factors. We group them by their underlying mechanism as shown in Table 7 in the Appendix of the main paper.

IA-1.1. Financing

CDI - Composite Debt Issuance

Following Lyandres *et al.* (2008), we compute composite debt issuance as the logarithmic growth rate of book debt (i.e., current debt, DLC, plus long-term debt, DLTT) over the last five years, i.e.,

$$\text{CDI}_t = \log(\text{DLC}_t + \text{DLTT}_t) - \log(\text{DLC}_{t-60} + \text{DLTT}_{t-60})$$

We replace the missing values of DLC and DLTT with zero individually. Additionally, the sum of DLC and DLTT must be larger than zero, and is set to missing otherwise.

CSI - Composite Share Issuance

Following Daniel and Titman (2006), we measure composite share issuance from CRSP data as the difference between the change in market equity (ME) (i.e., the shares outstanding, shrout, times the absolute value of their price, altprc, in million USD) and the cumulative log return (where r is a simple return) in each month from t to $t-60$, i.e.,

$$\text{CSI}_t = \log(ME_t/ME_{t-60}) - \sum_{\tau=t-59}^{t} \log(1+r_\tau)$$

DBE - Change in Common Equity

Following Richardson *et al.* (2005), we calculate the change in common equity as the change in common equity (CEQ) over one year scaled by lagged assets (AT), i.e.,

$$DBE_t = \frac{CEQ_t - CEQ_{t-12}}{AT_{t-12}}$$

DCOL - Change in Current Operating Liabilities

Following Richardson *et al.* (2005), we measure the change in current operating liabilities as the change in current operating liabilities (i.e., current liabilities, LCT, minus current debt, DLC) over one year scaled by lagged assets (AT), i.e.,

$$DCOL_t = \frac{(LCT_t - DLC_t) - (LCT_{t-12} - DLC_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC with 0.

DFNL - Change in Financial Liabilities

Following Richardson *et al.* (2005), we measure the change in financial liabilities (i.e., the sum of current debt, DLC, plus long-term debt, DLTT, plus preferred stocks, PSTK) over one year scaled by lagged assets (AT), i.e.,

$$DFNL_t = \frac{(DLC_t + DLTT_t + PSTK_t) - (DLC_{t-12} + DLTT_{t-12} + PSTK_{t-12})}{AT_{t-12}}$$

We replace the missing values of DLC, DLTT and PSTK with 0, if at least one of them is given.

NDF - Net Debt Financing

Following Bradshaw, Richardson, and Sloan (2006), we measure net debt financing as cash proceeds from the issuance of long-term debt (DLTIS) minus cash payments for long-term debt reductions (DLTR) plus net changes in current debt (DLCCH) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NDF_t = \frac{DLTIS_t - DLTR_t + DLCCH_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

We replace the missing values of DLCCH with zero.

NEF - Net Equity Financing

Following Bradshaw *et al.* (2006), we measure net equity financing as proceeds from the sale of common and preferred stocks (SSTK) minus payments for the repurchase of common and preferred stocks (PRSTKC) minus cash payments for dividends (DV) scaled by the average total assets (AT) over the current and the last year, i.e.,

$$NEF_t = \frac{SSTK_t - PRSTKC_t - DV_t}{\frac{1}{2}(AT_t + AT_{t-12})}$$

NXF - Net External Financing

Following Bradshaw *et al.* (2006), we measure net external financing as the sum of net debt financing (NDF) and net equity financing (NEF) as defined above.

IA-1.2. Intangibles

HR - Hiring Rate

Following Belo *et al.* (2014), we measure the hiring rate as the change in the number of employees (EMP) over one year scaled by the average number of employees, i.e.,

$$HR_t = \frac{EMP_t - EMP_{t-12}}{\frac{1}{2}(EMP_t + EMP_{t-12})}$$

We exclude firms with a hiring rate of zero.

OL - Operating Leverage

Following Novy-Marx (2011), we measure operating leverage as cost of goods sold (COGS) plus selling, general, and administrative expenses (XSGA), both scaled by current total assets (AT), i.e.,

$$OL_t = \frac{COGS_t + XSGA_t}{AT_t}$$

RER - Real-Estate Ratio

Following Tuzel (2010), we measure the real-estate ratio in two subperiods differently.

(a) Prior to 1983, the real-estate ratio is the sum of buildings (PPENB) and capital leases (PPENLS) scaled by net property, plant, and equipment (PPENT), i.e.,

$$RER_t = \frac{PPENB_t + PPENLS_t}{PPENT_t}$$

(b) After the end of 1983, the real-estate ratio is the sum of buildings at cost (FATB) and leases at cost (FATL), both divided by gross property, plant, and equipment (PPEGT), i.e.,

$$RER_t = \frac{FATB_t + FATL_t}{PPEGT_t}$$

Subsequently, we winsorize the real-estate ratios in each fiscal year at the first and 99th percentile. The industry-adjusted real-estate ratio is obtained by subtracting the industry average real-estate ratio from each stock-specific real-estate ratio. We use 2-digit SIC codes to assign stocks to industries. We always require at least five observations to calculate the industry average each year.

IA-1.3. Investment

ACI - Abnormal Corporate Investment

Following Titman *et al.* (2004), we measure abnormal corporate investments using the variable CE, which is capital expenditures (CAPX) scaled by sales (SALE), then ACI is given by CE scaled by CE from the previous three years minus one, i.e.,

$$ACI_t = \frac{CE_t}{\frac{1}{3}(CE_{t-12} + CE_{t-24} + CE_{t-36})} - 1$$

We follow Hou *et al.* (2020) and exclude stocks with sales below 10 million dollars.

AG - Asset Growth

Following Cooper *et al.* (2008), we measure asset growth as the change in total assets (AT) over one year scaled by lagged assets (AT), i.e.,

$$AG_t = \frac{AT_t - AT_{t-12}}{AT_{t-12}}$$

DINV - Inventory Changes

Following Thomas and Zhang (2002), we measure the change in inventory as the annual change in inventories (INVT) scaled by average total assets (AT) over the current and the last year, i.e.,

$$DINV_t = \frac{INVT_t - INVT_{t-12}}{\frac{1}{2}(AT_t + AT_{t-12})}$$

DNOA - Change in Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, the change in net operating assets is the difference between net operating assets of fiscal year t and fiscal year $t-12$ scaled by lagged assets (AT), i.e.,

$$DNOA_t = \frac{NOA_t^{int} - NOA_{t-12}^{int}}{AT_{t-12}}$$

DPIA - Change in Property, Plant, Equipment, and Inventory

Following Lyandres *et al.* (2008), we measure the change in property, plant, equipment, and inventory by combining the change in gross property, plant, and equipment (PPEGT) with the annual change in inventory (INVT) and scaling this sum by lagged total assets (AT), i.e.,

$$DPIA_t = \frac{(PPEGT_t - PPEGT_{t-12}) + (INVT_t - INVT_{t-12})}{AT_{t-12}}$$

DWC - Change in Net Non-Cash Working Capital

Following Richardson *et al.* (2005), we measure non-cash working capital as

$$WC_t = (ACT_t - CHE_t) - (LCT_t - DLC_t)$$

where ACT corresponds to current assets, CHE to cash, LCT to current liabilities, and DLC to short-term debt. We set missing values of DLC to zero. Then, the change in net non-cash

working capital corresponds to the change in WC from fiscal year $t-12$ to fiscal year t scaled by lagged total assets (AT), i.e.,

$$DWC_t = \frac{WC_t - WC_{t-12}}{AT_{t-12}}$$

IG - Investment Growth

Following Xing (2008), we measure investment growth as the annual change in capital expenditures (CAPX) scaled by lagged capital expenditures, i.e.,

$$IG_t = \frac{CAPX_t - CAPX_{t-12}}{CAPX_{t-12}}$$

NOA - Net Operating Assets

Following Hirshleifer *et al.* (2004), we measure net operating assets (defined as NOA^{int}) as

$$NOA_t^{int} = (AT_t - CHE_t) - (AT_t - DLC_t - DLTT_t - MIB_t - PSTK_t - CEQ_t)$$

where AT corresponds to total assets, CHE to cash and short-term investments, DLC to current debt, DLTT to long-term debt, MIB to minority interests, PSTK to the value of preferred stocks, and CEQ to common equity. Missing values in DLC, DLTT, MIB, and PSTK are set to zero. Then, net operating assets is the net operating assets of fiscal year t scaled by lagged assets (AT), i.e.,

$$NOA_t = \frac{NOA_t^{int}}{AT_{t-12}}$$

OA - Operating Accruals

Following Sloan (1996) (before 1988) and Hribar and Collins (2002) (after 1988), we measure operating accruals until 1988 as

$$OA_t = \frac{(\Delta ACT_t - \Delta CHE_t) - (\Delta LCT_t - \Delta DLC_t - \Delta TXP_t) - DP_t}{AT_{t-12}}$$

where ACT is current assets, CHE cash, LCT current liabilities, DLC short-term debt, TXP taxes payable, and DP depreciation and amortization. Moreover, we replace the missing values of DLC and TXP with zero. Δ represents changes over one year. From 1988 onwards, we measure operating accruals as

$$OA_t = \frac{NI_t - OANCF_t}{AT_{t-12}}$$

where NI is net income and OANCF corresponds to net cash flow from operations.

IA-1.4. Momentum

MOM - Return Momentum

Following Fama and French (1996), we measure return momentum from monthly CRSP returns (where r is a simple return) as the cumulative return from month $t-11$ to month $t-1$ skipping the most recent month, i.e.,

$$\text{MOM}_t = \exp\left(\sum_{\tau=t-11}^{t-1} \log(1+r_\tau)\right) - 1$$

RMOM - Residual Momentum

Following Blitz *et al.* (2011), we define the 11-month residual momentum (RMOM) in each month and for each stock as cumulative residual returns from month $t-11$ to month $t-1$, scaled by the standard deviation of residual returns over the same time horizon. Residual returns ϵ are obtained in each month from regressing monthly simple, excess stock returns from month $t-1$ to month $t-36$ on the Fama and French (1993) three-factor model. Throughout these rolling regressions, we always require 36 monthly returns.

RS - Revenue Surprise

Following Jegadeesh and Livnat (2006), we measure the change in revenues per share (DRPS) from quarterly Compustat data as quarterly sales (SALEQ) scaled by the product of quarterly shares outstanding (CSHPRQ) times their cumulative adjustment factor (AJEXQ), i.e.,

$$DRPS_q = \frac{SALEQ_q}{CSHPRQ_q \cdot AJEXQ_q} - \frac{SALEQ_{q-4}}{CSHPRQ_{q-4} \cdot AJEXQ_{q-4}}$$

Then, the revenue surprise corresponds to the change in revenues per share over the last four quarters scaled by the standard deviation of the change in revenues per share over the last eight quarters. We require at least six quarterly observations for this rolling standard deviation and remove infinite values. Finally, the earnings announcement date has to be after the fiscal quarter end date.[4]

SUE - Standardized Unexpected Earnings

Following Foster *et al.* (1984), we measure unexpected earnings per share (UES) from quarterly Compustat data as the change in split-adjusted earnings (which are equal to quarterly earnings per share, EPSPXQ, scaled by the correction factor for shares outstanding, AJEXQ) per share from its value four quarters ago

$$UES_q = \frac{EPSPXQ_q}{AJEXQ_q} - \frac{EPSPXQ_{q-4}}{AJEXQ_{q-4}}$$

Then, standardized unexpected earnings are defined as unexpected earnings per share divided by the standard deviation of unexpected earnings per share over the previous eight quarters.

[4]We follow the convention of Hou *et al.* (2020) and do not estimate a random walk model with drift to compute the surprise.

We require at least six quarterly observations for this rolling standard deviation. Finally, the earnings announcement date has to be after the fiscal quarter end date.[5]

52W - 52-Week High

Following George and Hwang (2004), we measure the 52-week high as month t's closing split-adjusted stock price scaled by the highest daily split-adjusted stock price over the previous twelve months. We only consider stock-month pairs with more than three valid price observations.

REV - Long-Term Reversal

Following De Bondt and Thaler (1985), we measure the long-term reversal effect by the cumulative returns from month $t-60$ to month $t-13$, i.e.,

$$\text{REV}_t = \exp\left(\sum_{\tau=t-60}^{t-13} \log(1+r_\tau)\right) - 1$$

IA-1.5. Profitability

CBOP - Cash-Based Operating Profitability

Following Ball *et al.* (2016), we measure cash-based operating profitability

$$\begin{aligned} CBOP_t = &(REVT_t - COGS_t - XSGA_t + XRD_t \\ &- \Delta RECT_t - \Delta INVT_t - \Delta XPP_t + \Delta DRC_t \\ &+ \Delta DRLT_t + \Delta AP_t + \Delta XACC_t)/AT_t \end{aligned}$$

where REVT is total revenue, COGS are cost of goods sold, XSGA are selling, general, and administrative expenses, and XRD are R&D expenses. Moreover, Δ refers to one-year changes, which we measure for accounts receivable (RECT), inventory (INVT), prepaid expenses (XPP), deferred revenues (the sum of current, DRC, and long-term, DRLT), trade accounts payable (AP), and accrued expenses (XACC). We set the missing values of XRD and all missing changes to zero.

GPA - Gross Profits to Assets

Following Novy-Marx (2013), we measure gross profits to assets as total revenues (REVT) minus cost of goods sold (COGS) scaled by current total assets (AT), i.e.,

$$GPA_t = \frac{REVT_t - COGS_t}{AT_t}$$

O - Ohlson's O-score

[5]We follow the convention of Hou *et al.* (2020) and do not estimate a random walk model with drift to compute the surprise.

Following Ohlson (1980), we measure O-score with the following linear relation

$$\begin{aligned} O_t = &- 1.32 - 0.407 \cdot \log(AT_t) + 6.03 \cdot \frac{DLC_t + DLTT_t}{AT_t} \\ &- 1.43 \cdot \frac{ACT_t - LCT_t}{AT_t} + 0.076 \cdot \frac{LCT_t}{AT_t} \\ &- 1.72 \cdot \mathbf{1}_{LT_t > AT_t} - 2.37 \cdot \frac{NI_t}{AT_t} - 1.83 \cdot \frac{PI_t + DP_t}{LT_t} \\ &+ 0.285 \cdot \mathbf{1}_{NI_t<0 \text{ and } NI_{t-12}<0} - 0.521 \cdot \frac{NI_t - NI_{t-12}}{|NI_t| + |NI_{t-12}|} \end{aligned}$$

where AT corresponds to total assets, DLC to short-term debt, DLTT to long-term debt, ACT to current assets, LCT to current liabilities, LT to total liabilities, PI to pretax income, DP to depreciation and amortization, and NI to net income. We winsorize all variables except for dummy variables at the first and 99th percentile of their respective distribution.

OPE - Operating Profits to Book Equity

Following Fama and French (2015), we measure operating profits to book equity as

$$OPE_t = \frac{REVT_t - COGS_t - XSGA_t - XINT_t}{BE_t}$$

where REVT corresponds to total revenues, COGS to cost of goods sold, XSGA to selling, general and administrative expenses, XINT to interest expenses, and BE to book equity. The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero. Moreover, missing values in COGS, XSGA, and XINT are set to zero.

ROA - Return on Assets

Following Balakrishnan *et al.* (2010), we measure return on assets as quarterly income before extraordinary items (IBQ) relative to lagged quarterly total assets (ATQ), i.e.,

$$ROA_q = \frac{IBQ_q}{ATQ_{q-1}}$$

Finally, the earnings announcement date has to be after the fiscal quarter end date.

ROE - Return on Equity

Following Hou *et al.* (2014), we measure return on equity as quarterly income before extraordinary items (IBQ) relative to lagged book equity (BEQ)

$$ROE_q = \frac{IBQ_q}{BEQ_{q-1}}$$

Quarterly book equity (BEQ) is computed as the (a) book equity of shareholders plus (b) balance sheet deferred taxes and investment tax credit minus (c) the book value of preferred stock. Regarding (a), we measure book equity of shareholders by shareholders' equity (SEQQ), or the sum of common equity (CEQQ) and the par value of preferred stock (PSTKQ), or if all previous items are unavailable by total assets (ATQ) minus total liabilities (LTQ). Regarding (b), the balance sheet deferred taxes and investment tax credit is TXDITCQ, or TXDBQ, or zero if both are missing. Regarding (c), the book value of preferred stocks corresponds to PSTKRQ, or PSTKQ, or zero if both are unavailable. Finally, the earnings announcement date has to be after the fiscal quarter end date.

TBI - Taxable Income to Book Income

Following Lev and Nissim (2004), we measure taxable income to book income as pretax income (PI) scaled by net income (NI)

$$TBI_t = \frac{PI_t}{NI_t}$$

We require positive pretax and net income.

IA-1.6. Size

ME - Size

Following Fama and French (1992), we measure size as the natural logarithm of the market equity. We obtain market equity data from CRSP by multiplying the shares outstanding (shrout * 1,000, as they are given in thousands) with the corresponding share price (the absolute value of altprc), i.e.,

$$ME_t = \log\left(shrout_t * 1000 * |altprc_t|\right)$$

IA-1.7. Trading frictions

AMI - Amihud Illiquidity Measure

Following Amihud (2002), we measure the illiquidity from daily return-to-volume (RV), which is the absolute daily return (r) scaled by the daily dollar trading volume (product of vol and prc), i.e.,

$$RV_t = \frac{|r_d|}{prc_t * vol_t}$$

Then, the Amihud illiquidity measure corresponds to the average return to volume estimate over the last six months. We require at least 50 observations for this average and adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

BETA - Beta Relative to the Market

Following Fama and MacBeth (1973), we measure the market beta β_1 from monthly CRSP data. Specifically, we run the following time series regression over the previous five years

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + u_t$$

Moreover, we require at least 24 monthly observations for the regression above.

DTV - Dollar Trading Volume

Following Brennan *et al.* (1998), we measure dollar trading volume from daily CRSP data as the average dollar trading volume (product of vol and prc) from month $t-6$ to month $t-1$ (with at least 50 observations). Moreover, we adjust dollar trading volume from NASDAQ according to Gao and Ritter (2010).

ISKEW - Idiosyncratic Skewness Relative to the Fama and French (1993) three-factor model

Following Bali *et al.* (2016), we regress daily excess returns in each month on the three Fama and French (1993) factor model (with a minimum of 15 daily observations), i.e.,

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

Then, idiosyncratic skewness is measured as the skewness of residuals u_t.

IVOL - Idiosyncratic Volatility Relative to the Fama and French (1993) model

Following Ang *et al.* (2006), we compute idiosyncratic volatility relative to the Fama and French (1993) three-factor model as the volatility of residuals from the following regression (with a minimum of 15 daily observations)

$$r_t^e = \alpha + \beta_1 \cdot (MKT_t - R_t^f) + \beta_2 \cdot SMB_t + \beta_3 \cdot HML_t + u_t$$

MDR - Maximum Daily Return

Following Bali *et al.* (2011), we measure the maximum daily (excess) return in each month from daily CRSP data (with at least 15 return observations).

TUR - Share Turnover

Following Datar *et al.* (1998), we measure the daily share turnover of each stock as the number of shares traded (vol) scaled by the number of shares outstanding (shrout), i.e.,

$$T_t = \frac{vol}{shrout}$$

Then, the share turnover for each firm in each month t is the average daily share turnover over the previous six months (with at least 50 observations). Moreover, we adjust the trading volume of NASDAQ stocks according to Gao and Ritter (2010).

IA-1.8. Valuation

BM - Book Equity to Market Equity

Following Davis, Fama, and French (2000), we measure the book-to-market ratio as book equity (BE) from Compustat divided by market equity (ME) from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$BM_t = \frac{BE_t}{ME_t}$$

The definition of book equity is (a) the book equity of shareholders plus (b) deferred taxes and investment tax credit (Compustat item TXDITC or TXDB + ITCB if TXDITC is unavailable) minus (c) the book value of preferred stock. Regarding (a), the book equity of shareholders is shareholders' equity (SEQ), or the sum of common equity (CEQ) and the par value of preferred stock (PSTK), or, if all previous items are unavailable, total assets (AT) minus total liabilities (LT). Regarding (b), deferred taxes (TXDITC) or investment tax credits (TXDB plus ITCB) if deferred taxes is unavailable. Regarding (c), the book value of preferred stock corresponds in the following order either to the redemption value (PSTKRV), or the liquidation value (PSTKL), or if all previous items are unavailable to the par value (PSTK). We replace missing values of items (b) and (c) with zero.

CFM - Cash Flow to Market Equity

Following Lakonishok *et al.* (1994), we measure the cash-flow-to-price ratio as income before extraordinary items (IB) plus depreciation (DP) scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$CFM_t = \frac{IB_t + DP_t}{ME_t}$$

We exclude all stocks with negative cash flows.

DM - Debt to Market Equity

Following Bhandari (1988), we measure the debt-to-market ratio as short-term debt (DLC) plus long-term debt (DLTT) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$DM_t = \frac{DLC_t + DLTT_t}{ME_t}$$

We exclude stocks with missing DLC and DLTT observations.

EBM - Enterprise Book Equity to Market Equity

Following Penman *et al.* (2007), we measure enterprise book equity scaled by market equity as net debt plus book equity scaled by net debt plus market equity

$$EBM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + BE_t}{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t + ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments, and ME to the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD). Lastly, in this case, book equity BE is computed as common equity (CEQ) plus TSTKP minus DVPA. Missing observations in DVPA and TSTKP are set to zero. Additionally, we require that the sum of net debt and book equity, as well as the sum of net debt plus market equity, are positive.

EM - Earnings to Market Equity

Following Basu (1983), we measure the earnings-to-price ratio as income before extraordinary items (IB) from Compustat divided by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$EM_t = \frac{IB_t}{ME_t}$$

We exclude firms with negative earnings.

NDM - Net Debt to Market Equity

Following Penman *et al.* (2007), we measure net debt to price as net debt from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$NDM_t = \frac{(DLTT_t + DLC_t + PSTK_t + DVPA_t - TSTKP_t) - CHE_t}{ME_t}$$

where DLTT corresponds to long-term debt, DLC to current debt, PSTK to the value of preferred stock, DVPA to preferred dividends in arrears, TSTKP to preferred treasury stock, CHE to cash and short-term investments. Lastly, missing observations in DVPA and TSTKP are set to zero.

NPY - Net Payout Yield

Following Boudoukh *et al.* (2007), we measure the net payout yield as net payouts from Compustat scaled by market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$\begin{aligned} NPY_t = &[(DVC_t + PRSTKC_t + \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV<0}) \\ &- (SSTK_t - \Delta PSTKRV_t \cdot 1_{\Delta PSTKRV>0})]/ME_t \end{aligned}$$

where DVC are dividends from common stock, PRSTKC is the purchase of common and preferred stock, PSTKRV is the value of the net number of preferred stocks outstanding, and SSTK reflects the sale of common and preferred stocks. $1_{\Delta PSTKRV<0}$ is a dummy variable that has value one if the annual change in PSTKRV is negative and zero otherwise. Moreover, we exclude stocks with negative net payouts.

OCM - Operating Cash Flow to Market Equity

Following Desai *et al.* (2004), we measure the ratio of operating cash flows to price as operating cash flows (OC) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$OCM_t = \frac{OC_t}{ME_t}$$

Before 1988, we measure OC as funds from operations (FOPT) minus the change in working capital (item WCAP). Thereafter, we measure OC as net cash flows from operating activities (OANCF). Moreover, we exclude firms with negative operating cash flows.

SM - Sales to Market Equity

Following Barbee Jr *et al.* (1996), we measure the sales-to-price ratio (SM) as sales (SALE) from Compustat divided by the market equity from CRSP (i.e., the fiscal-year end's shares outstanding, shrout, times the absolute value of their price, altprc, in million USD), i.e.,

$$SM_t = \frac{SALE_t}{ME_t}$$

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